Math 548

Portfolio Assignment 2

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Problem 1 (F22, P4)**.** Let $W \subset \mathbf{R}^5$ be the space spanned by the vectors

$$\left\{ \begin{pmatrix} 1 \\ -2 \\ 0 \\ 2 \\ -1 \end{pmatrix}, \begin{pmatrix} -2 \\ 4 \\ -1 \\ 1 \\ 2 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 2 \\ -2 \\ 1 \end{pmatrix} \right\}.$$

(a) Compute the dimension of W.

(b) Determine the dimension of $W^\perp$, and explain how this follows immediately from (a) using a theorem.

(c) Find a basis for $W^\perp$.

Proof. We will first put the following matrix into reduced row echelon form:

$$\begin{pmatrix} 1 & -2 & 0 & 2 & -1 \\ -2 & 4 & -1 & 1 & 2 \\ 0 & 1 & 2 & -2 & 1 \end{pmatrix} \xrightarrow{R_2 \to R_2 + 2R_1} \begin{pmatrix} 1 & -2 & 0 & 2 & -1 \\ 0 & 0 & -1 & 5 & 0 \\ 0 & 1 & 2 & -2 & 1 \end{pmatrix}$$

$$\xrightarrow{R_2 \leftrightarrow R_3} \begin{pmatrix} 1 & -2 & 0 & 2 & -1 \\ 0 & 1 & 2 & -2 & 1 \\ 0 & 0 & -1 & 5 & 0 \end{pmatrix}$$

$$\xrightarrow{R_3 \to -R_3} \begin{pmatrix} 1 & -2 & 0 & 2 & -1 \\ 0 & 1 & 2 & -2 & 1 \\ 0 & 0 & 1 & -5 & 0 \end{pmatrix}$$

$$\xrightarrow{R_2 \to R_2 - 2R_3} \begin{pmatrix} 1 & -2 & 0 & 2 & -1 \\ 0 & 1 & 0 & 8 & 1 \\ 0 & 0 & 1 & -5 & 0 \end{pmatrix}$$

$$\xrightarrow{R_1 \to R_1 + 2R_2} \begin{pmatrix} 1 & 0 & 0 & 18 & 1 \\ 0 & 1 & 0 & 8 & 1 \\ 0 & 0 & 1 & -5 & 0 \end{pmatrix}.$$

(a) Since there are three pivots, we have $\dim W = 3$.
(b) Since $\dim \mathbf{R}^5 = \dim W + \dim W^\perp$, we have $\dim W^\perp = 2$.

(c) Computing the nullspace of our matrix gives:

$$\begin{pmatrix}1 & 0 & 0 & 18 & 1\\ 0 & 1 & 0 & 8 & 1\\ 0 & 0 & 1 & -5 & 0\end{pmatrix}\begin{pmatrix}x_1\\ x_2\\ x_3\\ x_4\\ x_5\end{pmatrix} = \begin{pmatrix}0\\ 0\\ 0\\ 0\\ 0\end{pmatrix} = \begin{cases} x_1 = 18x_3 + x_5\\ x_2 = 8x_4 + x_5\\ x_3 = -5x_5\end{cases}$$

Taking x_4 and x_5 to be free, we get:

$$\mathcal{B}_{W^\perp} = \left\{ \begin{pmatrix}-18\\ -8\\ 5\\ 1\\ 0\end{pmatrix}, \begin{pmatrix}-1\\ -1\\ 0\\ 0\\ 1\end{pmatrix} \right\}.$$

□

Problem 2 (F19, P1)**.** Let P_3 be the real vector space of all real polynomials of degree three or less. Define $L : P_3 \longrightarrow P_3$ by $L(p(x)) = p(x) + p(-x)$.

(a) Prove that L is a linear transformation.

(b) Find a basis for the null space of L.

(c) Compute the dimension of the image of L.

Proof. (a) Let $p(x), q(x) \in P_3$ and $\alpha \in \mathbf{R}$. Observe that:

$$\begin{aligned} L(p(x) + \alpha q(x)) &= p(x) + \alpha q(x) + \big(p(-x) + \alpha q(-x)\big)\\ &= p(x) + p(-x) + \alpha(q(x) + q(-x))\\ &= L(p(x)) + \alpha L(q(x)). \end{aligned}$$

Thus L is a linear transformation.

(b) Let $\mathcal{B} = \{1, x, x^2, x^3\}$. Then:

$$\begin{aligned} L(1) &= 2\\ L(x) &= x + (-x) = 0\\ L(x^2) &= x^2 + (-x)^2 = 2x^2\\ L(x^3) &= x^3 + (-x)^3 = 0. \end{aligned}$$

This means:

$$[L]_{\mathcal{B}} = \begin{pmatrix}2 & 0 & 0 & 0\\ 0 & 0 & 0 & 0\\ 0 & 0 & 2 & 0\\ 0 & 0 & 0 & 0\end{pmatrix}.$$

Computing the nullspace of $[L]_{\mathcal{B}}$ gives:

$$\begin{pmatrix} 2 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}\begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \end{pmatrix} \implies \begin{cases} 2x_1 = 0 \\ 2x_3 = 0. \end{cases}$$

Hence $\mathcal{B}_{\ker[L]_{\mathcal{B}}} = \left\{\begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix}\right\}$.

(c) By the rank-nullity theorem, $\dim(\operatorname{im} L) = \dim(P_3) - \dim(\ker L) = 2$. □

Problem 3 (F18, P1)**.** Let $T : \mathbf{R}^3 \to \mathbf{R}^3$ be the linear transformation defined by $T\left(\begin{pmatrix} x \\ y \\ z \end{pmatrix}\right) = \begin{pmatrix} x + y \\ 2z - x \\ y + 2z \end{pmatrix}$.

(a) Find a matrix that represents T with respect to the standard basis for $\mathbf{R}^3$.

(b) Find a basis for the kernel of T.

(c) Determine the rank of T.

Proof. (a) Let $\mathcal{B}$ denote the standard basis in $\mathbf{R}^3$. From the following:

$$T\left(\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}\right) = \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix}$$
$$T\left(\begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}\right) = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}$$
$$T\left(\begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}\right) = \begin{pmatrix} 0 \\ 2 \\ 2 \end{pmatrix},$$

we get:

$$[T]_{\mathcal{B}} = \begin{pmatrix} 1 & 1 & 0 \\ -1 & 0 & 2 \\ 0 & 1 & 2 \end{pmatrix}.$$

(b) By row reducing $[T]_{\mathcal{B}}$:

$$\begin{pmatrix} 1 & 1 & 0 \\ -1 & 0 & 2 \\ 0 & 1 & 2 \end{pmatrix} \xrightarrow{R_2 \to R_2 + R_1} \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 2 \\ 0 & 1 & 2 \end{pmatrix}$$

$$\xrightarrow{R_3 \to R_3 - R_2} \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{pmatrix}$$

$$\xrightarrow{R_1 \to R_1 - R_2} \begin{pmatrix} 1 & 0 & -2 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{pmatrix},$$

we can see:

$$\begin{pmatrix} 1 & 0 & -2 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \implies \begin{cases} x_1 = 2x_3 \\ x_2 = -2x_3. \end{cases}$$

Taking x_3 to be free, a basis for $\ker T$ is $\left\{ \begin{pmatrix} 2 \\ -2 \\ 1 \end{pmatrix} \right\}$.

(c) The rank of a matrix is equal to the dimension of its image. By the rank-nullity theory, $\dim(\operatorname{im} T) = \dim(\mathbf{R}^3) - \dim(\ker T) = 2$. ☐

Problem 4 (S17, P1)**.** Let $V = \{a_0 + a_1\sqrt[3]{2} + a_2\sqrt[3]{4} \mid a_1, a_2, a_2 \in \mathbf{Q}\} \subseteq \mathbf{R}$. This is a vector space over $\mathbf{Q}$.

(a) Verify V is closed under product (using the usual product operation in $\mathbf{R}$).

(b) Let $T : V \to V$ be the linear transformation defined by $T(v) = (\sqrt[3]{2} + \sqrt[3]{4})v$. Find the matrix that represents T with respect to the basis $\{1, \sqrt[3]{2}, \sqrt[3]{4}\}$ for V.

(c) Determine the characteristic polynomial for T.

Proof. (a) Let $a_0 + a_1\sqrt[3]{2} + a_2\sqrt[3]{4}$ and $b_0 + b_1\sqrt[3]{2} + b_2\sqrt[3]{4}$ be elements of V. We have:

$$\begin{aligned}(a_0 + a_1\sqrt[3]{2} + a_2\sqrt[3]{4})&(b_0 + b_1\sqrt[3]{2} + b_2\sqrt[3]{4}) \\ &= (a_0b_0 + 2a_1b_2 + 2a_2b_1) + \sqrt[3]{2}(a_0b_1 + a_1b_0 + 2a_2b_2) + \sqrt[3]{4}(a_0b_2 + a_1b_1 + a_2b_0).\end{aligned}$$

Thus V is closed under product.

(b) Let $\mathcal{B} = \{1, \sqrt[3]{2}, \sqrt[3]{4}\}$. Then:

$$\begin{aligned} T(1) &= \sqrt[3]{2} + \sqrt[3]{4} \\ T(\sqrt[3]{2}) &= 2 + \sqrt[3]{4} \\ T(\sqrt[3]{4}) &= 2 + 2\sqrt[3]{2}. \end{aligned}$$

This means:

$$[T]_{\mathcal{B}} = \begin{pmatrix} 0 & 2 & 2 \\ 1 & 0 & 2 \\ 1 & 1 & 0 \end{pmatrix}.$$

(c) The characteristic polynomial of $[T]_{\mathcal{B}}$ is:

$$\begin{aligned} \det\big([T]_{\mathcal{B}} - \lambda I\big) &= \det\begin{pmatrix} -\lambda & 2 & 2 \\ 1 & -\lambda & 2 \\ 1 & 1 & -\lambda \end{pmatrix} \\ &= -\lambda \det\begin{pmatrix} -\lambda & 2 \\ 1 & -\lambda \end{pmatrix} - 2\det\begin{pmatrix} 1 & 2 \\ 1 & -\lambda \end{pmatrix} + 2\det\begin{pmatrix} 1 & -\lambda \\ 1 & 1 \end{pmatrix} \\ &= -\lambda(\lambda^2 - 2) - 2(-\lambda - 2) + 2(1 + \lambda) \\ &= -\lambda^3 + 6\lambda + 6. \end{aligned}$$

□

Problem 5 (F19, P2)**.** Let $T : \mathbf{R}^3 \to \mathbf{R}^3$ be the linear transformation that rotates counterclockwise around the z-axis by $\frac{2\pi}{3}$.

(a) Write the matrix for T with respect to the standard basis $\left\{\begin{pmatrix}1\\0\\0\end{pmatrix}, \begin{pmatrix}0\\1\\0\end{pmatrix}, \begin{pmatrix}0\\0\\1\end{pmatrix}\right\}$.

(b) Write the matrix for T with respect to the basis $\left\{\begin{pmatrix}\frac{\sqrt{3}}{2}\\-\frac{1}{2}\\0\end{pmatrix}, \begin{pmatrix}0\\1\\0\end{pmatrix}, \begin{pmatrix}0\\0\\1\end{pmatrix}\right\}$.

(c) Determine all (complex) eigenvalues of T.

(d) Is T diagonalizable over $\mathbf{C}$? Justify your answer.

Proof. (a) Let $\mathcal{E}$ denote the standard basis. We have:

$$\begin{aligned} T\left(\begin{pmatrix}1\\0\\0\end{pmatrix}\right) &= \begin{pmatrix}-\frac{1}{2}\\\frac{\sqrt{3}}{2}\\0\end{pmatrix}, \\ T\left(\begin{pmatrix}0\\1\\0\end{pmatrix}\right) &= \begin{pmatrix}-\frac{\sqrt{3}}{2}\\-\frac{1}{2}\\0\end{pmatrix}, \\ T\left(\begin{pmatrix}0\\0\\1\end{pmatrix}\right) &= \begin{pmatrix}0\\0\\1\end{pmatrix}. \end{aligned}$$

Hence:

$$[T]_{\mathcal{E}} = \begin{pmatrix} -\frac{1}{2} & -\frac{\sqrt{3}}{2} & 0 \\ \frac{\sqrt{3}}{2} & -\frac{1}{2} & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

(b) Let $\mathcal{B} = \left\{ \begin{pmatrix} \frac{\sqrt{3}}{2} \\ -\frac{1}{2} \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \right\}$. We have:

$$\begin{aligned} T\left(\begin{pmatrix} \frac{\sqrt{3}}{2} \\ -\frac{1}{2} \\ 0 \end{pmatrix} \right) &= \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \\ T\left(\begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} \right) &= \begin{pmatrix} -\frac{\sqrt{3}}{2} \\ -\frac{1}{2} \\ 0 \end{pmatrix}, \\ T\left(\begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \right) &= \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}. \end{aligned}$$

Hence:

$$[T]_{\mathcal{B}} = \begin{pmatrix} 0 & -1 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

(c) We must first find the characteristic polynomial of $[T]_{\mathcal{B}}$:

$$\begin{aligned} \det\big([T]_{\mathcal{B}} - \lambda I\big) &= \det \begin{pmatrix} -\lambda & -1 & 0 \\ 1 & 1-\lambda & 0 \\ 0 & 0 & 1-\lambda \end{pmatrix} \\ &= (1-\lambda) \det \begin{pmatrix} -\lambda & -1 \\ 1 & 1-\lambda \end{pmatrix} \\ &= (1-\lambda)(\lambda^2 - \lambda + 1). \end{aligned}$$

Hence $\lambda = 1$, $\lambda = \frac{1}{2} + \frac{\sqrt{3}}{2}i$, and $\lambda = \frac{1}{2} - \frac{\sqrt{3}}{2}i$ are the eigenvalues of $[T]_{\mathcal{B}}$.

(d) Since T has three distinct eigenvalues, and since $\dim(\mathbf{R}^3) = 3$, T is diagonalizable over $\mathbf{C}$.

$\square$